AGRICULTURAL SUPPORT TRIAGE AGENT

Course Name: AGENTIC AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1.	DEEPAK PATRO	EN22CS301316
2.	BHOOMIKA PATIDAR	EN22CS301270
3.	CHARU NARIYA	EN22CS301290
4.	DEV KUMAR DAS	EN22CS301321
5.	AVICHAL MISHRA	EN22CS301243

Group Name:07D3

Project Number:AAI-31

Industry Mentor Name:

University Mentor Name:Prof. Maya Yadav

Academic Year:2025-2026

PROBLEM STATEMENT AND OBJECTIVES

1. PROBLEM STATEMENT

Agriculture remains the primary livelihood for a large portion of India's population, yet farmers often lack timely access to reliable, scientific, and government-verified information. Farmers typically face challenges such as:

• Crop selection uncertainty
• Sudden crop diseases and pest attacks
• Lack of awareness about government schemes and subsidies
• Fragmented information spread across multiple websites
• Delay in expert consultation in rural areas

Traditional advisory systems rely on manual consultation, which is time-consuming, expensive, and not always accessible to small and marginal farmers.

There is a strong need for an **intelligent, automated, and easily accessible advisory system** that can provide real-time, reliable, and domain-specific farming guidance.

2. PROBLEM OBJECTIVES

The main objectives of this project are:

1. Build an **AI powered farming assistant** that answers farmer queries instantly.
2. Design a **multi-agent AI system** where each agent specializes in a specific agricultural domain.
3. Use **Retrieval Augmented Generation (RAG)** to ensure responses are based on trusted agricultural knowledge.
4. Provide **disease detection + treatment suggestions** for urgent crop issues.
5. Provide **crop advisory** for farming best practices.
6. Provide **government policy & scheme awareness**.
7. Automatically **classify farmer query urgency**.
8. Build a **modern user interface** for easy farmer interaction.
9. Generate a **downloadable agriculture report** for users.

3. SCOPE OF PROJECT

The system is designed to assist farmers with **text-based agricultural queries** through an AI interface.

INCLUDED IN SCOPE:

• Crop advisory and farming practices
• Crop disease identification and treatment suggestions
• Government schemes and subsidy awareness

• Knowledge retrieval from agriculture websites
• AI multi-agent coordination
• Web-based UI using Streamlit

OUT OF SCOPE:

• Real image-based disease detection
• IoT sensor integration
• Real-time weather API integration
• Regional language support (future scope)

PROPOSED SOLUTION

This project proposes a **Multi-Agent AI Advisory System** powered by:

• CrewAI for agent orchestration
• Groq LLM (Llama-3.1-8B) for fast reasoning
• RAG pipeline for trusted knowledge retrieval
• Streamlit for user interface

The system uses **four specialized AI agents**:

Agent	Responsibility
Priority Classifier	Detect urgency level
Crop Advisor	Crop planning & farming advice
Disease Expert	Disease detection & treatment
Policy Advisor	Government schemes

Each agent collaborates sequentially to generate a **complete agriculture report**.

1. KEY FEATURES

• Multi-Agent AI architecture
• Retrieval Augmented Generation (RAG)
• Automated knowledge scraping from FAO & Government sites
• Intelligent query urgency classification
• Modern web interface
• Downloadable farming report
• Fast inference using Groq LLM
• Scalable modular architecture

2. OVERALL ARCHITECTURE / WORKFLOW

2.1 Data Collection Layer:

A custom web scraper collects agriculture knowledge from:

• FAO agriculture pages
• FAO climate change pages
• Indian Government Agriculture Welfare website

The scraper extracts:
• Headings
• Paragraphs
• Bullet points

The data is stored in:

```
data/agriculture_docs.txt
```

This forms the **knowledge base**.

2.2 RAG Pipeline (Knowledge Engine):

The RAG pipeline converts the collected data into searchable embeddings.

Steps:

1. Load text using LangChain TextLoader
2. Split into chunks (800 characters, overlap 150)
3. Convert chunks to embeddings using **sentence-transformers/all-MiniLM-L6-v2**
4. Store embeddings in **Chroma Vector Database**
5. Create a retriever to fetch top relevant knowledge.

This allows agents to answer using **trusted agriculture data**.

2.3 Multi-Agent AI System (Core Intelligence)

CrewAI orchestrates four domain experts:

2.3.1 PRIORITY AGENT

Classifies farmer query as:
• Urgent
• Normal
• Planning

2.3.2 CROP ADVISOR AGENT

Provides:
• Crop selection

• Farming practices
• Yield improvement advice

2.3.3 DISEASE EXPERT AGENT

Provides:
• Symptom analysis
• Disease identification
• Treatment suggestions

2.3.4 POLICY ADVISOR AGENT

Provides:
• Government schemes
• Subsidies
• Insurance programs

Agents collaborate through **sequential task execution**.

2.4 Streamlit User Interface

The UI provides:

Sidebar:
• Crop selection
• Region selection
• Priority selection

Main Panel:
• Query input box
• Generate AI advice button
• Results dashboard
• Download report feature

2.5 End-to-End Workflow

1. Farmer enters query
2. CrewAI triggers multi-agent pipeline
3. Priority agent classifies urgency
4. RAG tool retrieves knowledge
5. Each expert agent generates response
6. Results displayed in dashboard
7. Report available for download

3. TOOLS AND TECHNOLOGIES

Category	Tools
LLM	Groq Llama-3.1-8B
Multi-Agent Framework	CrewAI
RAG Framework	LangChain
Embeddings	Sentence Transformers
Vector Database	ChromaDB
Web Scraping	BeautifulSoup
UI	Streamlit
Programming Language	Python
Deployment Ready	Yes

RESULTS AND OUTPUT

1. SYSTEM OUTPUT

The system generates a **complete AI Agriculture Report** containing:

1.Detected Query Priority
2.Crop Farming Advice
3.Disease Identification & Treatment
4.Government Scheme Suggestions

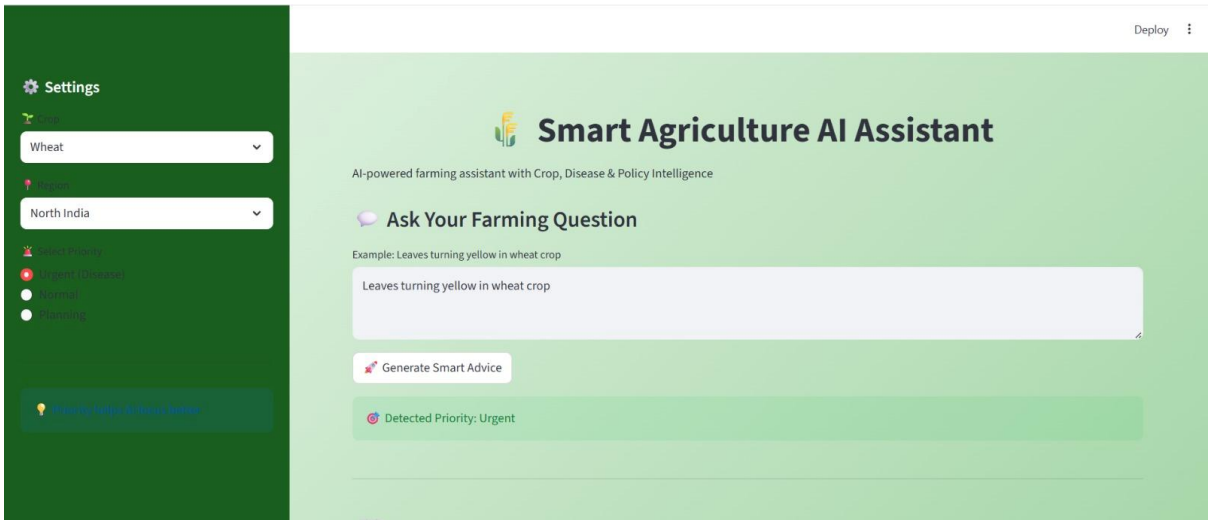
2. SCRRENSHOTS OF WORKING PROJECT

2.1 HOME SCREEN – USER INPUT INTERFACE

The following screenshot shows the main interface of the Smart Agriculture AI Assistant developed using Streamlit.
This screen allows farmers to:

• Select crop type
• Select farming region
• Select priority level (Urgent / Normal / Planning)
• Enter farming questions
• Generate AI powered advice

The system provides a simple and user-friendly interface designed for non-technical users such as farmers.



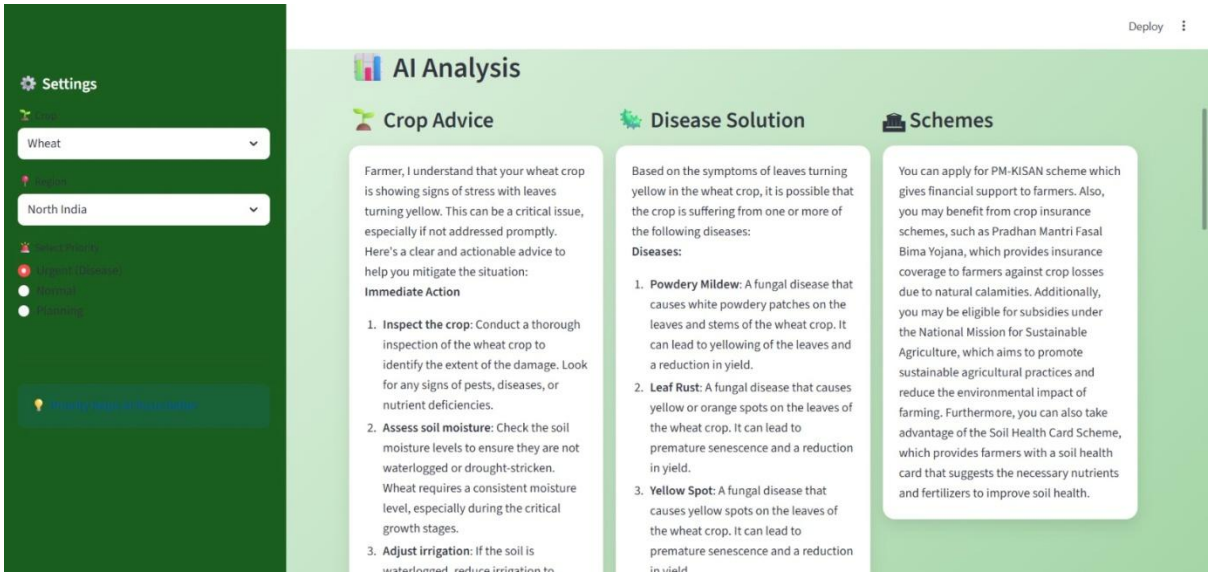
2.2 AI GENERATED AGRICULTURAL ANALYSIS DASHBOARD

This screenshot shows the AI generated analysis produced by the multi-agent system after processing the farmer query.

The dashboard displays results from multiple specialized AI agents:

• Crop Advisor → Provides farming recommendations
• Disease Expert → Detects diseases and treatments
• Policy Advisor → Suggests government schemes
• Priority Agent → Detects urgency of the query

This demonstrates successful integration of:
• CrewAI Multi-Agent System
• RAG Knowledge Retrieval
• Groq LLM Inference



2. KEY OUTCOMES

• Built a fully functional **multi-agent AI system**
• Successfully integrated **RAG with CrewAI**
• Achieved **domain-specific knowledge grounded responses**
• Created **end-to-end working product**
• Designed scalable modular architecture

CONCLUSION

The Smart Agriculture AI Assistant demonstrates how modern Generative AI and Multi-Agent Systems can revolutionize agriculture advisory services.

The project successfully combines:
• Multi-Agent AI reasoning
• Retrieval Augmented Generation
• Real agricultural knowledge sources
• User-friendly interface

This system can significantly improve farmers' access to timely and reliable agricultural guidance, helping increase productivity, reduce crop loss, and improve awareness of government support programs.

The project also provided deep practical exposure to:
• LLM integration
• Agent orchestration
• RAG pipelines
• AI product development

FUTURE SCOPE AND ENHANCEMENT

Future improvements can include:

• Image-based crop disease detection
• Regional language support (Hindi, Gujarati, etc.)
• Real-time weather API integration
• Mobile app deployment
• Voice input support
• Farmer profile personalization
• Integration with soil & IoT sensors